

Multi Wall Polycarbonate Sheet Product

8mm Twinwall Polycarbonate Glazing

SPEC SECTION 08-80-00

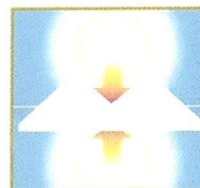


Resolite®

A DIVISION OF STABILIT AMERICA



• 2 Side UV Protection • Heat Reflective



Colors Available

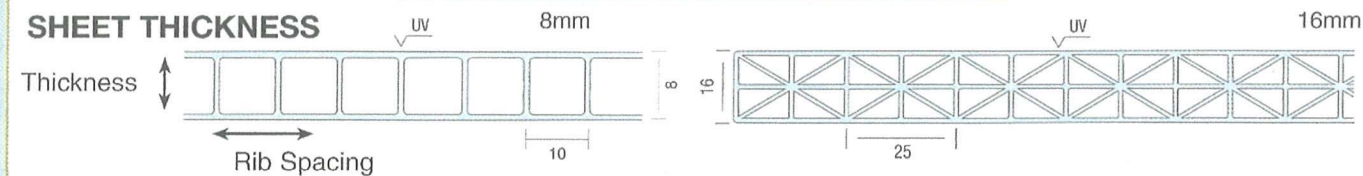
- High Diffused HD Soft White
- Clear
- Opal
- Bronze
- Grey
- Blackout

Makrolon®

10 & 15 YEAR NON PRORATED WARRANTY AVAILABLE

Makrolon®

POLYCARBONATE MULTI WALL SHEET



Width (in.)	47.25", 48", 71.25", 72" and 82.67"
Length (ft.)	Up to 48'
Gauges	4mm(3/16"), 6mm(1/4"), 8mm(5/16"), 10mm(3/8"), 16mm(5/8")

MAKROLON Product Range

Sheet Thickness	4mm	6mm	8mm	8mm	10mm	16mm
Sheet Type	Twin	Twin	Twin	Triple	Twin	X Wall
Weight (lb/ft²)	0.16	0.26	0.31 0.35	0.31	0.35	0.51
Weight (g/m²)	800	1300	1500 1700	1500	1700	2500
Light Transmission (%) Clear	82	82	81	72	80	59
Light Transmission (%) Bronze	52	50	50	NA	46	31
Light Transmission (%) Opal	61	55	54	N/A	54	54
Light Transmission (%) HD Soft White	N/A	N/A	77	68	N/A	55
Light Transmission (%) Light Deprivation Black and White	N/A	N/A	0	N/A	N/A	0

TECHNICAL DATA Product Property Table

Property/Thickness	Unit	4mm	6mm	8mm Twin	8mm Triple	10mm	16mm
Thermal Insulation BTU/hr-ft²-°F	U-Value R-Value	0.72 1.39	0.62 1.62	0.58 1.73	.54 1.86	0.53 1.89	0.42 2.40
Cold Forming Radius (Minimum Permissible)	Inch	29.5	39.4	49.2	51.18	59.1	94.5
Heat Expansion Coefficient	Inch/ft°F	4x10 ⁻⁴	4x10 ⁻⁴	4x10 ⁻⁴	4x10 ⁻⁴	4x10 ⁻⁴	4x10 ⁻⁴
Max. Permanent Service Temperature w/o Load	°F	212	212	212	212	212	212
Possible Expansion Due to Heat and Moisture	Inch/Inch	0.003	0.003	0.003	0.003	0.003	0.003
Weighted Sound Reduction Index	db	N/A	10	14	14	16	21

Special Features Available:

No Drip – anti condensate that inhibits the formation of condensation into droplets by reducing surface tension making the condensation run off the sheet into the gutters or drain.

IQ Relax – offers high thermal insulation without sacrificing natural light. The Makrolon panels are coextruded with an advanced additive technology that adapts to weather conditions which helps maintain a comfortable temperature inside the structure. During summer months IQ Relax reflects heat and solar energy while in winter it keeps the heat in.

IQ Select – designed for retail garden centers offering high light transmission while reflecting the solar energy making the temperature of the shopping environment very comfortable.

Light Deprivation Panel - 0% light transmission multi wall polycarbonate sheet with the option of black, white or combination of both colors. This sheet is designed for the end and side walls of a greenhouse which requires a light deprivation system.



Standard H-Profile



Standard U-Profile

Multi Wall Accessories

H-Channel UV Protected	24' length
U-Channel UV Protected	12' length
Vented and Non Vented Tape	150' roll



Loading Guidelines:

For a 4' wide sheet:

Maximum Purlin Spacing (Inches) Sheet Deflection - 1"				
Load (lbs/ft ²)	15	30	45	60
6mm	31"	24"	12"	*
8mm	38"	28"	18"	10"
10mm	45"	34"	28"	24"
16mm	75"	43"	36"	30"

For a 6' wide sheet:

Maximum Purlin Spacing (Inches) Sheet Deflection - 1"				
Load (lbs/ft ²)	15	30	45	60
6mm	25"	9"	*	*
8mm	36"	26"	12"	*
10mm	44"	33"	27"	20"
16mm	60"	40"	36"	30"

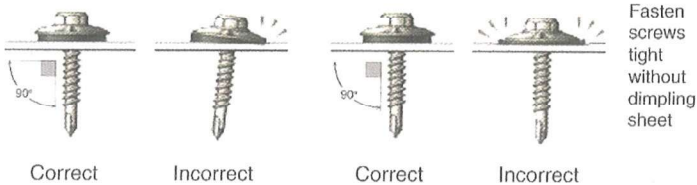
Polycarbonate Guide for Cutting, Storage and Installation

Cutting: To cut sheet width and length use circular saw and fine tooth plywood blade (at least 10 teeth per inch). Cut sheets before taking off protective poly film. Clean sheet flutes out after cutting with vacuum or blow out channels with an air compressor before installation.

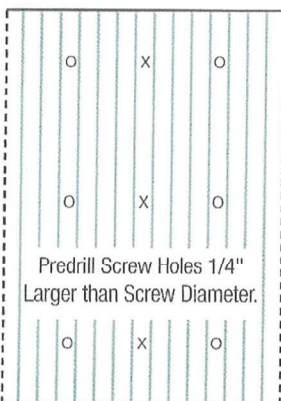
Storage: Store sheets out of direct sunlight in a dry area. Keep sheets in a well ventilated location.

Thermal Expansion: For each linear foot, allow 1/32" for expansion or contraction of sheet for every 100 degree temperature differential (contact Resolite for the expansion rate on colored materials). Pre-drill screw holes 1/4" larger than screw diameter.

Screw Attachment



Pre-drill Screw Holes 1/4"
Larger than Screw Diameter.



Sheet Fastening
Positions

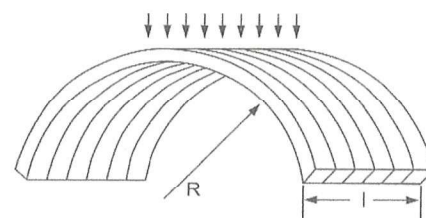
X = Mid-Sheet
fasteners
for 48" sheet

O = Mid-Sheet
fastener
for 72" sheet

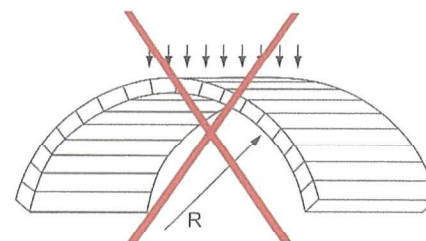
Fasten sheets in
all four corners

Sheet Orientation:

Ribs should always run vertical



Vertical

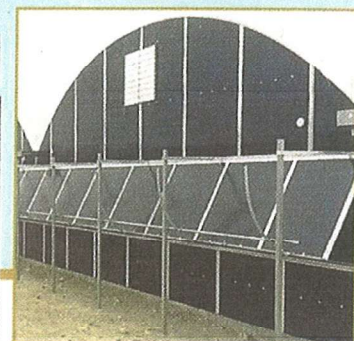


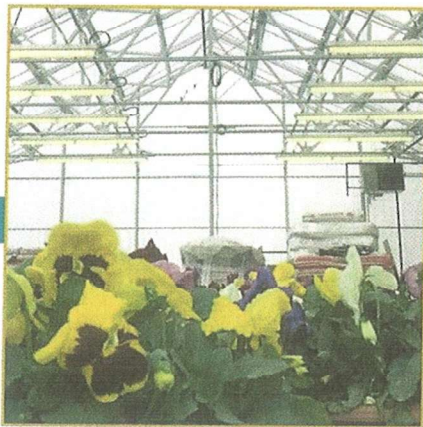
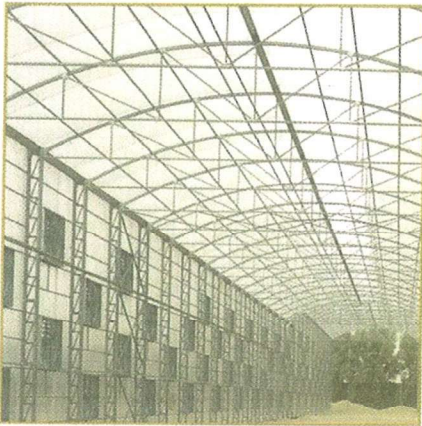
Horizontal



Fire Rating: 8mm

ASTM	CLASSIFICATION
ASTM E84-01	Class A (NFPA); Class I (UBC)
ASTM D635-74	Class CC1
ASTM D2843-93	Low Smoke Density (UBC standard No. 26.5)
ASTM D1929-68	Self Ignition Temperature greater than 650° F





Corrugated Polycarbonate also available

Makrolon®

10 & 15 Year Non Prorated Warranty

Makrolon Multi Wall sheets are available in both a 10 & 15 Year Non Prorated Warranty against loss of light transmission caused by yellowing and breakage caused by hail.

The polycarbonate panels come with U.V. coextruded protection on the exterior surface. This protection will prevent the sheets from discoloration and degradation causing the sheet to become brittle and lose light transmission. Two sided co-extruded UV protection is available as well as the "No Drip" anti condensate feature.

Virtually unbreakable, the Makrolon Multi Wall sheets made with Makrolon resin will withstand breakage caused by hail stones of up to 20mm in diameter for the life of the warranty.

Makrolon PC greenhouse covering prevents the transmission of over 99.9% of harmful radiation resulting from UV rays. It maintains its mechanical and structural integrity under extreme weather conditions (up to 212° F).

Special Notes:

1. The installation must comply with the IMSA Plastics guidelines for warranty to be valid. Local authorization may be required.
2. Installation guidelines available on request.

For additional information or questions contact Resolite at 888-RESOLITE (737-6548).



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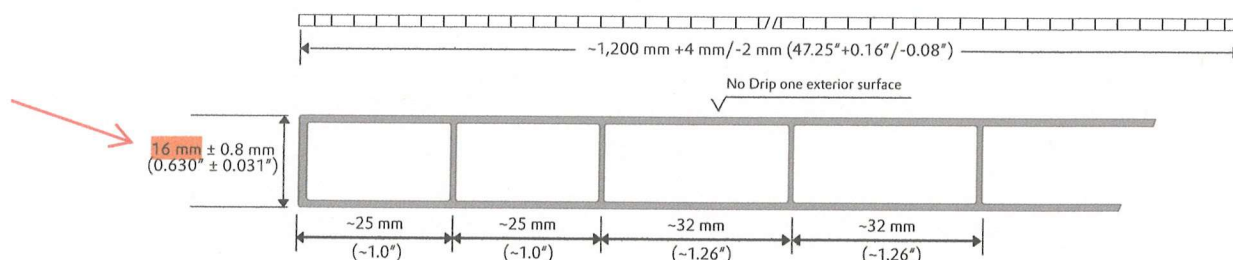


POPC-FLY2-RE-0214-N



Makrolon® is a registered trademark of Covestro.

16 mm ACRYLITE® Resist high impact acrylic double-skin sheet



16 mm ACRYLITE® Resist high impact acrylic double-skin sheet is a light transmitting, heat-insulating and weather-resistant sheet made of impact-modified acrylic (polymethyl methacrylate, PMMA) polymer.

16 mm ACRYLITE® Resist high impact acrylic double-skin sheet is suitable for sloped, curved and vertical applications including the following:

- Canopies
- Carports
- Clerestory Glazing
- Interior Walls
- Interior Partitions
- Lighting Panels
- Conservatories
- Skylights
- Barrel Vaults
- Facades & Cladding
- Greenhouses
- Pool Enclosures
- Patio Covers
- Sunrooms

Advantages of 16 mm ACRYLITE® Resist high impact acrylic double-skin sheet

- Extraordinary clear sheet is UV-resistant and includes a 30-year warranty against yellowing.
- Is coated with a completely non-toxic biologically neutral Anti Algae Application
- Offers excellent protection against excessive amounts of UV radiation.

- Up to 50% energy savings and minimizes CO² emissions.
- Hail-resistant with a 10-year guarantee for withstanding hailstones with a kinetic energy of 1 joule.
- High light transmission in the PAR and visible spectral energy wavelengths.
- Impact resistance during transportation, handling and installation.
- Available in clear and a range of colors and solar tints.

Warranties

Non-prorated, full replacement 30 year non-yellowing, 10 year light transmission and 10 year hail warranties. For details see warranty.

Environmental Sustainability

ACRYLITE® Resist high impact acrylic double-skin sheets' natural heat insulating qualities can translate into significant energy savings, making them an ideal choice for eco-lighting and building green. The sheets are built to last using environmentally sound manufacturing processes in facilities that have received ISO-14001 environmental certification. In addition, if the time does come for replacement, ACRYLITE® can be recycled in an environmentally friendly manner.

Technical Data (Typical values)

	Light Transmittance D65	Total Energy Transmission Values	Shading Coefficient (SC)
Clear (ORS22)	86%	82%	0.94
Light White (WRS22)	74%	73%	0.83
Bronze (8R22)	53%	64%	0.75
Solar Cool White (WRS23)	37%	46%	0.53
UV Transmission	blocking		
U-value	2.5 W/m²K (0.49 BTU/hr•ft²•°F)		
R-value	2.04 °F/BTU•hr•ft²		
Coefficient of Heat Expansion (α)	0.09 mm/m °C (0.00005 in/in/°F)		
Expansion Due to Heat and Moisture	6 mm/m (1/16"/ft)		
Thickness	16 mm (5/8")		
Rib Spacing	32 mm (1 1/4")		
Width	1200 mm (47 1/4")		
Length	up to 10973 mm (36 ft)		
Approximate Area Weight	4.5 Kgm² (0.9 lb/ft²)		
Weighted Sound Reduction Index	24 dB		
Maximum Service Temperature	70°C (160 °F)		
ASTM D-635 (Rate of Burn)	C2 / CC2		
ASTM D-1929 (Self Ignition Temp)	830 °F		
ASTM D-2843 (Smoke Density Rating)	7.0 %		
CAN/ULC S102.2	< 150 Flame Spread Classification		
DIN 4102	normal combustability, B2		
Minimum permissible cold-forming radius of 96" (150 x thickness)			
Values are approximate.			

No Drip

The water-dispersing, patented No Drip coating applied on one side of the sheet causes any surface water to form a thin, continuous film. When the sheet is installed with the No Drip coating on the outside of the roof, the coating can support the natural cleaning of the roof by rainwater. When the coating is on the inside, it contributes toward preventing any condensed water from forming drops and thus averts the damage caused by condensation.

Anti Algae Coating

The Anti Algae coating, based on advanced nanotechnology, is a further development of the successful no drip coating. Due to the coating, algae, mosses and pollen lose the ability to adhere to the sheet and may be immediately and completely decomposed by the natural UV radiation from the sun.

Fire Behaviour

- The fire behavior of ACRYLITE® is rated as C2 or CC2 according to ASTM D-635.
- ACRYLITE® burns almost entirely without smoke according to DIN4102 and ASTM D-2843 and is easily extinguished.
- The smoke gases produced by ACRYLITE® are neither acutely toxic according to DIN 53436 nor corrosive according to DIN VDE 0482-267.

Load Bearing Capacity

Due to its excellent rigidity, large areas can be glazed quickly and efficiently. Few intermediate supports are required to carry substantial uniformly distributed loads (refer to Support Spacing data chart). Reduction of structural members means less shading thus increasing light levels.

Support Spacing

As flat glazing supported on all sides, 1200 mm wide 16 mm double-skin sheet requires no additional cross members for uniformly distributed loads up to 15.7 lb/ft² (750 N/m²). For greater loads, the recommended spacing is given in the support spacing chart. Maximum allowable spacing is in the direction parallel to the sheet's ribs. Refer to local building codes to determine the applicability of these values to specific applications. Building codes will indicate the design loads to be used to determine the maximum span lengths or support spacing.

Support Spacing Chart

16 mm ACRYLITE® Resist high impact acrylic double-skin sheet width as delivered 1200 mm (47 1/4")			
Load		Support Spacing	
[N/m ²]	(lb/ft ²)	[m]	(in)
750	15.7		
1000	20.9	4.0	157
1250	26.1	3.1	122
1500	31.3	2.8	110
1750	35.5	2.6	102
2000	41.8	2.5	98
2250	47	2.4	94
2500	52.2	2.2	86
2750	57.4	2.1	82
3000	62.6	2.0	78

ACRYLITE® is a registered trademark of Evonik Cyro LLC in the Americas. These same products are manufactured and marketed under the PLEXIGLAS® trade name on the European, Asian, African and Australian continents.

Certified to DIN EN ISO 9001 (Quality) and DIN EN ISO 14001 (Environment).

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